Smart AI-Powered Library Management and Recommendation System

Agile Project Management Plan

Group T | 4-Week Timeline

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1. Main goal

Develop a smart web-based library system with book management, borrow/return functions, AI chatbot, recommendation features, and admin dashboard.

2. Team roles and responsibilities

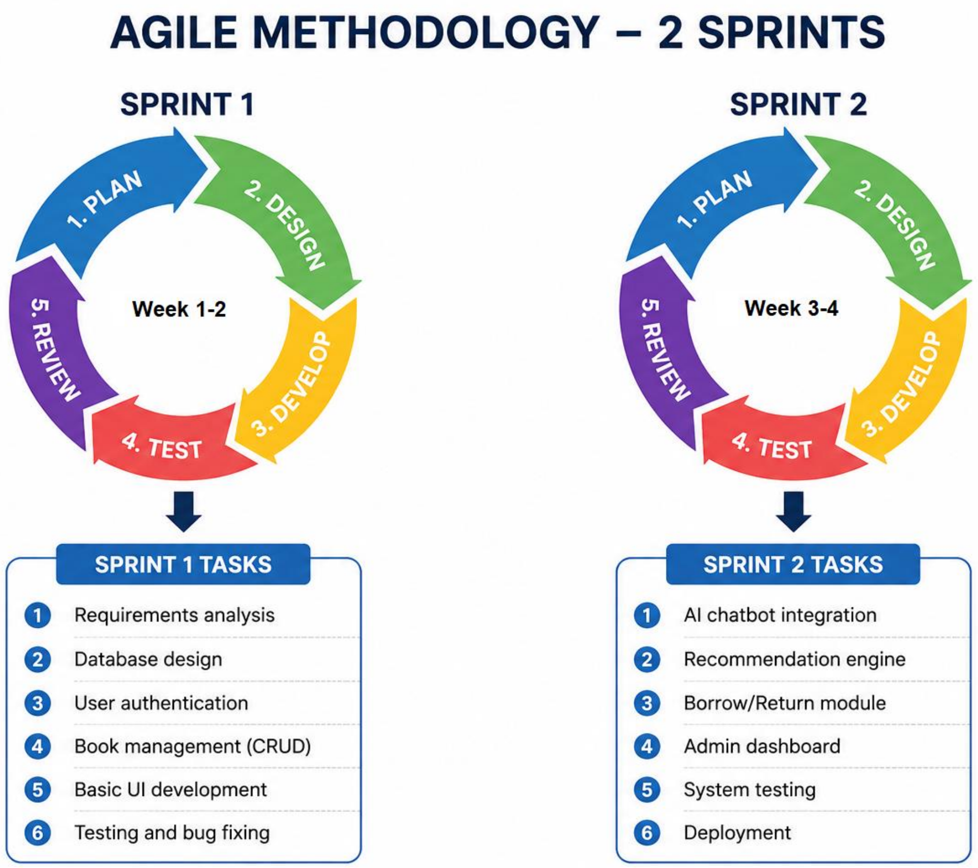
Team member	Main role	Responsibilities
Sangam Poudel	Frontend and UI support	UI pages, Bootstrap layout, user forms, book search pages, dashboard UI, testing support.
Batbold Chuluunbaatar	Backend, database, and AI support	Python backend, SQLite database, APIs, authentication, AI chatbot, recommendation logic, deployment support.
Both members	Shared project responsibilities	Requirements analysis, planning, integration, testing, sprint reviews, retrospectives, report, and presentation.

3. Sprint schedule

Sprint	Timeline	Duration	Main focus	Expected result
Sprint 1	Week 1-2	2 weeks	Requirements, design, database setup, core UI, authentication, basic book management.	Basic system structure and core library functions are ready.
Sprint 2	Week 3-4	2 weeks	Borrow/return functions, AI features, admin dashboard, testing, deployment, final documentation.	Final integrated system is tested, deployed, and ready for presentation.

4. Product backlog

ID	Backlog item	Description	Priority	Sprint
P1	User authentication	Login, register, and role-based access for students, librarians, and administrators.	High	Sprint 1
P2	Book management	Add, update, delete, search, and view book records.	High	Sprint 1
P3	Database design	Create tables for users, books, loans, reviews, and reservations.	High	Sprint 1
P4	Borrow and return module	Allow users to borrow and return books and update available copies.	High	Sprint 2
P5	AI chatbot assistant	Answer user questions and help users search for books.	Medium	Sprint 2
P6	Recommendation engine	Recommend books based on genre, borrowing history, ratings, or simple rules.	Medium	Sprint 2
P7	AI book summaries	Generate short book summaries and suggested audience information.	Medium	Sprint 2
P8	Admin insights dashboard	Show active users, popular books, borrowing trends, and late return patterns.	Medium	Sprint 2
P9	Testing and bug fixing	Test main user flows and fix identified problems.	High	Sprint 2
P10	Deployment and final documentation	Deploy application and prepare report/presentation materials.	High	Sprint 2



5. Sprint 1

Sprint 1 timeline: Week 1-2 (2 weeks)

Sprint 1 goal: Build the foundation of the system, including requirements, design, database setup, authentication, and basic book management.

- Finalize project requirements and confirm the scope for a 4-week project.
- Design the system architecture, database structure, and key user interface pages.
- Set up the backend project, frontend structure, and database connection.
- Develop authentication and basic book management features.

Task ID	Task	Owner	Timeline	Deliverable
1.1	Review and finalize project requirements	Both members	Week 1	Completed requirement list
1.2	Create system architecture and module plan	Both members	Week 1	Architecture and module design
1.3	Design database tables for users, books, loans, and reviews	Member 2	Week 1	Database design
1.4	Create UI wireframes for login, dashboard, book list, and book form	Member 1	Week 1	UI wireframes
1.5	Set up Python Flask/Django project and folder structure	Member 2	Week 1-2	Backend project structure
1.6	Build frontend layout using HTML, CSS, Bootstrap, and JavaScript	Member 1	Week 1-2	Reusable UI layout
1.7	Implement login/register and role-based access	Member 2	Week 2	Authentication module
1.8	Implement basic book add, update, delete, search, and list functions	Both members	Week 2	Book management module
1.9	Sprint 1 testing and bug fixing	Both members	End of Week 2	Tested Sprint 1 features

Deliverable	Description
Requirement specification	Finalized list of functional and non-functional requirements.
System design	Architecture, database structure, UI wireframes, and module plan.
Database setup	Initial SQLite database with core tables.
Authentication module	Users can register and log in with basic role support.
Book management module	Basic CRUD and search functions are implemented.

6. Sprint 2

Sprint 2 timeline: Week 3-4 (2 weeks)

Sprint 2 goal: Complete the main system functions, integrate AI features, test the system, deploy the project, and prepare final documentation.

- Complete borrowing and returning book functionality.
- Integrate chatbot, recommendations, AI summaries, and smart search features.
- Develop admin dashboard and reporting features.
- Test all major user flows and fix bugs.
- Deploy the system and prepare final report and presentation.

Task ID	Task	Owner	Timeline	Deliverable
2.1	Develop borrow and return module	Both members	Week 3	Borrow/return workflow
2.2	Update available copies after borrowing and returning	Member 2	Week 3	Database update logic
2.3	Develop review and rating functions	Member 1	Week 3	Review/rating interface
2.4	Integrate AI chatbot assistant	Member 2	Week 3	Chatbot feature
2.5	Add smart search and recommendation engine	Both members	Week 3-4	AI search and recommendation feature
2.6	Add AI-generated book summary feature	Member 2	Week 4	Summary feature
2.7	Build admin insights dashboard	Member 1	Week 4	Dashboard page
2.8	Perform unit, integration, UI, and AI output testing	Both members	Week 4	Test results and bug list
2.9	Deploy application using Docker/cloud hosting	Both members	Week 4	Deployed application
2.10	Prepare final report and presentation	Both members	Week 4	Final submission documents

Deliverable	Description
Borrow and return module	Users can borrow and return books, and available copy count is updated.
AI chatbot assistant	Users can ask questions and receive guidance about library use and book search.
Recommendation engine	System recommends books using borrowing history, category, rating, or rule-based logic.
AI summaries and smart search	Users receive short book summaries and can search using natural language.
Admin dashboard	Administrators and librarians can view important borrowing and activity insights.
Final tested system	Main features are tested and bugs are fixed.
Deployment and documentation	Project is deployed and final report/presentation is prepared.